

Structural Primers Trigger Procedural Shortcuts in LLMs: Answer Leakage, Invariants and Feature Resolution on Graphs

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Abstract

Graph neural networks routinely receive structural encodings such as node degree, clustering and random-walk return probabilities as input features. We serialize the same descriptors as a textual *primer* before a graph given to a language model. Does the primer change how the model executes a graph algorithm, and which structural distinctions survive serialization? We evaluate Qwen3-1.7B and Qwen3-4B, with and without thinking, on 40-node Erdős–Rényi graphs at seven edge densities ($p=.10$ to $.85$), over 84,000 generations with gold answers verified against reconstructed graphs. Two controls separate access to an answer from its use: a graph-blind solver flags primers that leak the answer, and a paired diagnostic checks a model’s degree and neighborhood answers for one node against each other. A degree primer lifts plain Qwen3-4B’s edge counting from 2.00% to 29.25%, yet lowers its joint degree–neighborhood correctness from 85.75% to 76.50%. On degree queries the same model skips the neighbor list in 44–74% of responses. It gains 27 points at $p=.85$, where its own count is 33% accurate, and loses up to 12 where the count is 99% accurate. Qwen3-1.7B with thinking instead keeps counting and uses the primer as a check (+16.0 points). Printed to two decimals, random-walk features take on average 3.30 distinct values across a graph’s 40 nodes at $p=.50$, fewer than degree alone. A leaked answer can change which procedure a model runs, and its effect follows the accuracy of the procedure it displaces. Neither route is evidence of graph understanding.

1 Introduction

A language model (LLM) reads a graph given as text token by token. To answer “what is the degree of node 16?” it must find node 16’s adjacency line and count it. [Fatemi et al. \(2024\)](#) showed that accuracy on such GraphQA tasks depends strongly on the textual encoding. Graph learning has a standard

remedy for a model that cannot easily compute a structural quantity: supply it. Degree, clustering and random-walk structural encodings (RWSE) are input features of graph neural networks (GNNs) and graph transformers ([Dwivedi et al., 2022](#); [Rampásek et al., 2022](#)). We study the textual analogue, a *structural primer*: one sentence per node, placed before the graph, stating its degree, clustering coefficient or RWSE.

Research question. *How do textual structural primers alter an LLM’s procedural execution of graph tasks, and what are the expressivity limits of these primers in preserving structural invariants?* Aggregate accuracy cannot answer it: it conflates *access* to a text feature that already states the answer, *execution* of the algorithm the task requires, and *consistency* with relations any correct answer must satisfy. A primer that states every degree turns node_degree into lookup, so a gain there says nothing about whether the model counts. A model that looks up one answer and counts another can violate the identity $d_v=|N(v)|$, which holds on every simple graph. Nor does accuracy show whether a printed feature still separates the nodes its exact values separate, which bounds what a primer can convey at all.

Contributions. (1) A *graph-blind solver* that reads only the rendered primer and node/edge counts and marks where a primer leaks the answer (§3.3). (2) A *paired consistency diagnostic* that scores a model’s degree and neighborhood answers for the same node against $d_v=|N(v)|$, separating agreement (C) from joint correctness (J) (§3.5). (3) A classification of responses into retrieval and enumeration, which shows the primer replacing or checking the counting procedure (§4.2). (4) A count of the distinct values a two-decimal RWSE primer takes per graph, which from $p=.35$ falls below that of degree (§4.5).

2 Related Work

Graphs as text. GraphQA (Fatemi et al., 2024), NLGraph (Wang et al., 2023), GPT4Graph (Guo et al., 2023) and GraphArena (Tang et al., 2025) vary encodings, prompts and tasks; NLGift (Zhang et al., 2024) asks whether gains from tuning on synthetic graph data generalize. Sanford et al. (2024) show that a single-layer transformer can represent node degree and edge count, suggesting that difficulty lies in execution, not architectural capacity. Huang et al. (2024) find that neighborhood text helps node classification through label-relevant phrases rather than structure. Dziri et al. (2023) show that transformers solve compositional tasks through shortcut subgraphs of the computation. Graph-QA evaluations mostly score final answers. We know of none that records the procedure behind an answer or checks two answers about one node against each other.

Structural encodings. Message-passing GNNs are bounded by the 1-WL test (Xu et al., 2019), whose first refinement round assigns each node its degree. RWSE and related encodings add information beyond that bound (Dwivedi et al., 2022; Rampásek et al., 2022). A GNN receives them as real-valued vectors, in which small differences between nodes of equal degree stay distinct. Graph-Token (Perozzi et al., 2024) instead learns a graph encoder for a frozen LLM. A textual primer needs no training, but it passes every value through the tokenizer as a fixed-precision digit string. We know of no measurement of how much node-level resolution this serialization keeps.

Shortcuts and partial inputs. Shortcut learning (Geirhos et al., 2020) and hypothesis-only baselines (Gururangan et al., 2018; Poliak et al., 2018) show that high accuracy can come from cues that do not require the intended computation. A partial input’s failure does not rule a shortcut out (Feng et al., 2019). We apply the partial-input idea to an intervention (the primer) rather than a dataset. Added text can itself degrade reasoning (Shi et al., 2023; Levy et al., 2024), so we include a content-free length control.

3 Method

3.1 Corpus and models

Graphs are Erdős–Rényi $G(n, p)$ (Erdős and Rényi, 1959; Gilbert, 1959) with $n=40$, 100 independently

Condition	States	Chars	deg.	e.cnt
none	no primer	–	14.4	3.0
filler	“Node i is simply present”	1,829	14.4	3.0
components	component count (graph)	37	14.4	3.0
clustering	clustering coefficient	1,629	14.4	10.4
rwse	return prob., 2 and 3 steps	2,949	21.0	3.0
degree	degree	891	100	100
all	the three rows above	4,691	100	100

Table 1: Conditions (per node unless marked), mean length, and graph-blind solver accuracy (%) on node_degree (deg.) and edge_count (e.cnt), over seven densities, 300 held-out graphs each. **Bold:** answer-carrying. No condition exceeds 0.81% on connected_nodes or 74.6% on edge_existence.

seeded graphs per density. The main sweep covers $p \in \{.10, .20, .35, .50\}$, six GraphQA tasks and seven conditions (67,200 generations). An extension adds $p \in \{.65, .75, .85\}$ for node_degree and edge_existence only (16,800). node_degree, connected_nodes and edge_existence query the same node of a graph. Four arms answer every prompt: Qwen3-1.7B and Qwen3-4B (Qwen Team, 2025) with thinking off (1.7B, 4B) and on (1.7B-T, 4B-T). Decoding is greedy with an 8,192-token budget (2,048 for the plain arms in the extension). Every saved gold answer matches the graph rebuilt from its seed; the 84,000 unique generations form 840 cells of 100. Every graph here has 40 nodes and a cycle, so node_count and cycle_check have a constant gold and are excluded from comparisons.

3.2 Primers

A prompt consists of the primer, the graph in the incident encoding of Fatemi et al. (2024), and the question. All seven conditions share one renderer (Table 1). Clustering and RWSE, the return probabilities (P_{vv}^2, P_{vv}^3) of the random walk $P=D^{-1}A$ (0 for isolated nodes), are printed to two decimals. filler restates the node set (“Node i is simply present within the graph G ”), which the encoding already lists, at 12% above the length of clustering.

3.3 Graph-blind solver

The solver reads the rendered primer text and the graph’s node and edge counts (except the count a task asks for), never the edge list. It applies 16 exact rules where they exist, and otherwise one heuristic or 8 rules fitted on graphs disjoint from those scored. Its accuracy is therefore a lower

bound on what the primer and counts determine. A primer is *answer-carrying* on a task when the solver scores at least 85% there, and *side information* otherwise. Any threshold in $(74.6, 100]$ gives the same split. Only degree and all carry answers, on `node_degree` (the queried degree) and `edge_count` ($|E| = \frac{1}{2} \sum_v d_v$).

3.4 Scoring and inference

The primary outcome is $S = \mathbf{1}[\hat{y}=y] \cdot \mathbf{1}[\text{not capped}]$: exact match (set equality for neighbors), with a generation that reaches its budget scored as a failure. Main-sweep effects pool the four densities ($n=400$ paired graphs). Other effects and all response shares (§4.2, §4.4) use terminated generations, dropping a pair when either generation is capped; this coincides with S wherever nothing is capped, as for plain 4B on `node_degree`. Effects are paired differences in percentage points (pp) against none, with exact McNemar tests (McNemar, 1947) and Benjamini–Hochberg adjustment (Benjamini and Hochberg, 1995) over the six primer contrasts per (arm, task), or (arm, metric) for C and J . Intervals are 95% paired bootstrap intervals over graphs, stratified by density (Efron, 1979). All tests are exploratory; intervals are not simultaneous.

3.5 Cross-query consistency

For node v let $\hat{d}_v$ and $\hat{N}_v$ be the model’s answers to the separate `node_degree` and `connected_nodes` prompts. $C_v=1$ when both complete, $\hat{d}_v$ parses as an integer, the predicted neighbors are valid node ids, and $\hat{d}_v=|\hat{N}_v|$. In that case the two answers satisfy the degree identity. $J_v=S_{v,\text{deg}} S_{v,\text{nbr}}$ is joint correctness. Since $J_v \leq C_v$, $C-J$ is the rate of *consistent but wrong* pairs. C tests a necessary relational invariant at the output level, not an internal graph representation.

3.6 Procedure classification

An *enumerating* `node_degree` response restates the queried node’s neighbor list and itemizes at least five entries, one per line. A *retrieving* response answers without restating the list. A retrieval need not return the stated degree; accuracy among retrievals measures how often it does. A thinking response *reports a discrepancy* when it uses a discrepancy word (discrepancy, contradiction, conflict, inconsistency) outside a hypothetical. Without a primer there is nothing to retrieve, so

Arm	Task	None	Fill.	Comp.	Clust.	RWSE	Deg.	All
1.7B	Deg	60.50	−0.50	1.75	3.00	4.00	7.50*	10.25*
	Nbr	72.50	−6.25*	0.75	1.25	1.00	3.50	2.00
	EE	70.50	−15.75*	−8.25*	3.75	5.00	4.00	11.50*
	EC	1.25	−0.50	−1.00	−0.50	−0.75	3.00	0.75
1.7B-T	Deg	76.25	−3.00	−1.00	1.50	2.50	8.75*	12.25*
	Nbr	78.75	2.50	2.50	2.00	0.25	6.25*	6.25*
	EE	99.00	−0.50	−0.75	−1.00	−1.25	−2.50	−1.75
	EC	15.75	−1.50	−0.75	−1.25	−3.25	−3.25	5.00
4B	Deg	99.25	−0.50	0.25	−0.50	−2.25*	−6.50*	−8.00*
	Nbr	86.00	−12.50*	9.25*	1.00	0.00	−3.00	−2.75
	EE	90.50	−7.00*	−2.75	4.00*	−10.00*	−4.75*	−10.00*
	EC	2.00	1.25	−1.25	−1.25	−1.50	27.25*	3.75*
4B-T	Deg	97.00	2.00	0.25	1.75	−1.25	2.50	0.50
	Nbr	95.50	1.25	0.50	−0.75	0.50	0.50	−1.25
	EE	99.75	0.00	−0.50	−0.75	−1.25	0.00	−1.00
	EC	20.00	12.25*	24.75*	24.75*	10.25*	8.50*	1.75

Table 2: Main sweep ($p \leq .50$, $n=400$ per cell): success S (%; capped = 0) under none and paired effects (pp) on `node_degree` (Deg), `connected_nodes` (Nbr), `edge_existence` (EE) and `edge_count` (EC). *: BH-adjusted $q < .05$.

accuracy under none is the accuracy of the model’s own count.

4 Results

4.1 Answer access is not answer use

Table 2 shows that an answer-carrying primer does not reliably raise accuracy on the task it leaks. degree reduces `edge_count` to a 40-term sum. Plain 4B sets out to sum degrees in 96–100% of responses even without a primer. With degree it goes from 2.00% to 29.25% (+27.25 [22.75, 31.75]; 114 graphs fixed, 5 broken). Plain 1.7B gains only +3.00 ($q=.101$). Its sums move closer (mean absolute error 136 to 40 edges, terminated generations) but are rarely exact. 1.7B-T changes by −3.25. The thinking arms rarely complete an edge count. Without a primer 4B-T finishes 20.50% of generations and is correct on 97.56% of those, for 20.00% success. Under degree it completes 41.25% but succeeds on only 28.50%. components and clustering carry no edge count (solver 3.0% and 10.4%), yet both reach 44.75%, 12.50 pp above filler (32.25%; intervals [6.75, 18.01] and [7.50, 17.75]). components states one component for every graph at $p \geq .20$, and there it still beats filler (64/35/8% against 46/22/1% at $p=.20/.35/.50$). Its gain acts mainly on whether the thinking arm completes (45.50% against 33.50% under filler), not on graph-specific content.

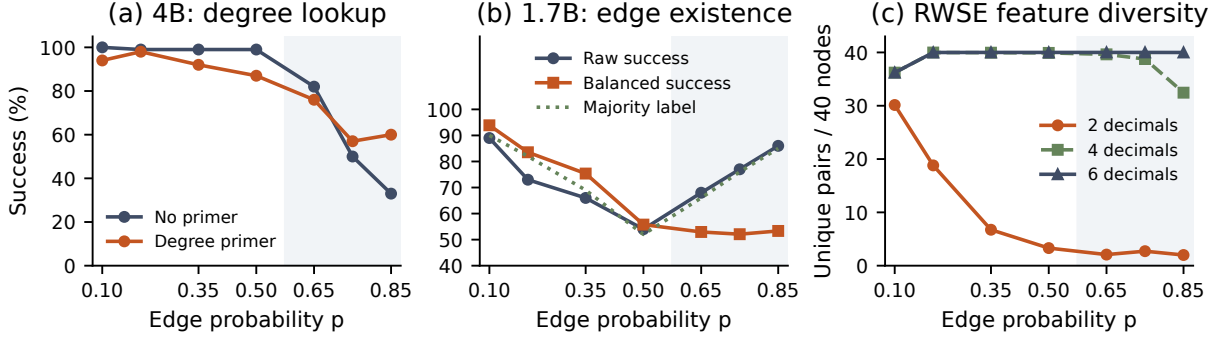


Figure 1: Density diagnostics (100 graphs per point; shading: the $p \geq .65$ extension, plain-arm budget 2,048 tokens). (a) Plain 4B node_degree success, none and degree. (b) Plain 1.7B edge_existence, no primer: raw, balanced and majority-label success. (c) Mean distinct RWSE pairs among a graph’s 40 nodes at 2, 4 or 6 decimals; models saw two decimals.

4.2 Primers replace or check the counting procedure

Table 3 follows plain 4B on node_degree across all seven densities. Without a primer its count is 99–100% accurate up to $p=.50$ and falls to 33% at .85. Under degree it often skips the neighbor list, retrieving in 44–74% of responses at 73–98% accuracy. At $p=.35-.75$, enumeration falls from 85–100% of responses to at most 8%. The primer’s effect follows the accuracy of the count it replaces: -6 , -1 , -7 , -12 and -6 pp at $p=.10-.65$, then $+7$ and $+27$ at .75 and .85 (100 graphs each; Figure 1a). At $p=.50$ the primer turns 13 correct answers into errors and corrects one. filler, twice as long, costs 2 pp, so the loss is not a cost of added text. From $p=.50$ the responses that do not retrieve are only 79%, 45%, 28% and 16% accurate: 68–92% of them open with an answer, usually not the stated degree, and then recount. Under all, which states the degree among two other statistics, retrieval falls to 15–51% and accuracy drops at every density, by up to 37 pp at $p=.65$.

The same primer induces a different procedure in 1.7B-T. It retrieves in at most 2% of responses and, from $p=.35$, enumerates in 76–98%. It uses a discrepancy word in 23% of responses up to $p=.50$ and 49% beyond (0.3% and 4% without a primer), and is correct on 61% and 69% of those. Over the seven densities degree adds +16.0 pp [12.0, 19.7] with capped generations scored as errors (+19.9 [16.3, 23.6] over terminated pairs). It also raises capping from 1.1% to 7.6%. Plain 1.7B retrieves in at most 4% of responses. It gains +10 and +18 pp only at $p=.20$ and .35, where its own count is 80% and 41% accurate. From $p=.50$, with baselines of 5–30%, it stays within 2 pp. As a replacement,

p	.10	.20	.35	.50	.65	.75	.85
acc., none	100	99	99	99	82	50	33
enumerates	1	34	95	100	94	85	44
acc., degree	94	98	92	87	76	57	60
retrieves	44	56	74	62	71	64	69
acc. retrievals	89	98	91	92	89	73	80
acc. the rest	98	98	96	79	45	28	16
enumerates	2	16	6	8	2	1	0

Table 3: Plain Qwen3-4B on node_degree by density (% , 100 graphs each, none capped). Procedures as in §3.6; the five-entry rule undercounts enumeration at $p \leq .20$. The rest: responses that do not retrieve.

the leaked answer helps or hurts according to the accuracy of the count it displaces. As a check on a count 1.7B-T still executes, it helps.

4.3 Agreement does not certify correctness

For plain 4B (Table 4), degree lowers J from 85.75% to 76.50% (-9.25 [-13.75 , -4.75]; 28 nodes fixed, 65 broken) and C from 87.00% to 81.00%. The degree task carries the larger share (-6.50 , against -3.00 on neighbors). The answer-carrying primer thus buys a leaked global gain on edge_count while local joint correctness falls, though by no more than under the length control. filler lowers J by 13.25 pp, through the neighborhood task (-12.50). components, a 37-character sentence, raises J to 95.00% ($+9.25$ [6.00, 12.75]). For plain 1.7B, degree raises J by 8.50 pp [4.25, 12.75], but C rises further, to 73.00% against J of 62.50%. So 10.50% of pairs satisfy $d_v=|N(v)|$ with at least one wrong answer: agreement overstates correctness. For 4B-T, whose J is 93.75% without a primer, degree adds 2.00 [-1.25 , 5.00].

Arm	None	Degree	All
1.7B	62.25/54.00	73.00/62.50	68.00/61.75
1.7B-T	81.00/71.00	81.00/76.50	83.50/79.50
4B	87.00/85.75	81.00/76.50	80.00/77.50
4B-T	94.00/93.75	96.00/95.75	93.75/93.25

Table 4: Degree–neighborhood agreement / joint correctness (C/J , %; 400 queried nodes per cell, $p \leq .50$).

4.4 Side information acts through global cues

On edge_existence the share of queried pairs that are edges rises with density, from 10% at $p=.10$ to 85% at .85. Pooled over densities, the plain arms answer “yes” to 97–100% of true edges under every condition, and 94–99% of their errors are false alarms. Plain 1.7B’s success without a primer rises from 54% at $p=.50$ to 86% at .85, but it answers “yes” to 94% and 99% of queries there. Its balanced success is 55.77% and 53.33%; always answering “yes” scores 85% at .85 (Figure 1b). Primers move this label bias rather than edge recall. Pooled over the seven densities, all lowers 1.7B’s false-alarm rate from 51% to 33% (−18.6 [−24.0, −13.1]). filler, which states nothing about edges, raises it to 69% (+18.3 [13.9, 22.7]). Where counting degrades, clustering adds 11.3 pp [6.7, 16.3] to plain 4B on node_degree above $p=.50$. The model keeps enumerating (69% of responses, against 74% without a primer). The printed coefficients barely vary within a graph (mean within-graph SD at most 0.021), so any gain rests at most on a graph-level value. On these cells degree, clustering and rwse gain 9.3, 11.3 and 7.7 pp, but all, which combines them, loses 21.3.

4.5 Two decimals collapse RWSE resolution

In a GNN, RWSE is useful partly because its continuous values can separate nodes of equal degree, which the first 1-WL round cannot tell apart. In a prompt, each value becomes a fixed-precision digit string, and values that round alike become identical token strings. Printed to two decimals, the (P_{vv}^2, P_{vv}^3) pairs of the 40 nodes take on average 30.14 distinct values at $p=.10$, 6.74 at .35, 3.30 at .50 and 2.00 at .85 (Figure 1c). At $p=.50$ the most common pair is shared by 62.68% of nodes. From $p=.35$ the degree primer takes more distinct values than the RWSE primer (11.98 against 6.74 at $p=.35$; 12.50 against 3.30 at .50). There, the printed encoding separates fewer node classes than the first 1-WL round, although RWSE is meant to add information beyond 1-WL. The loss occurs at

the printed precision, not in the graph: at six decimals the same features take 40.00 distinct values at $p=.50$ and 39.99 at .85. Without a higher-precision condition, we attribute no accuracy change to the loss.

5 Discussion and Conclusion

Where a primer carries the answer, a model can change procedure, and the sign of the effect depends on what is displaced. Where traceable, side information acts via completion and “yes” bias. Agreement overstates correctness, and a leaked answer can lower joint correctness while raising a global count. From $p=.35$, two-decimal RWSE separates fewer node classes than 1-WL’s first round.

For graph ML evaluation, gains from a structural interface should be reported with (i) a graph-blind score, to separate leakage from use; (ii) per-cell baselines, on which a leaked answer’s effect depends; (iii) joint correctness and $C-J$, not agreement alone; (iv) balanced accuracy where density changes label prevalence; (v) a content-free length control; and (vi) the number of distinct printed feature values per graph. On this corpus, structural primers trigger procedural shortcuts that help where the model’s own count partly fails, harm where the count succeeds, and act as a check only for a model that still counts. None of these routes shows general graph understanding.

Limitations

We evaluate two sizes of one model family on synthetic Erdős–Rényi graphs, with one encoding, node order, primer position and prompt template. Node position matters: retrieval is more accurate for nodes 0–9 (96% against 83%) (Liu et al., 2024; Ge et al., 2025). Decoding is greedy. The extension’s 2,048-token plain-arm budget confounds density with budget above $p=.50$, and the thinking arms’ completion effects may be specific to our budgets. Per-density effects rest on 100 graphs each. filler is an approximate, not token-matched, length control. Procedures are classified by surface patterns, for plain 4B, 1.7B and 1.7B-T only, and a model may follow a procedure it does not state (Turpin et al., 2023). A targeted, non-exhaustive audit of the reused answer extractor (52 responses) moves C and J by at most 1.25 pp per cell. We test graph-query primitives, not node classification or link prediction.

Ethics Statement

Open-weight models on synthetic graphs; no human subjects, personal data or annotation.

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